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# John Jones' construction of manifold in dimension 30 with Kervaire invariant one

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## Notation:

- $\mathbb{Z}_2 := \mathbb{Z}/2\mathbb{Z}$
- $\Sigma_t$ : the group of permutations of a set with  $t$  elements
- Denote  $\Sigma = \{\sigma, 1\}$ , where  $\sigma$  is the nontrivial element.
- Let  $\tau_Y$  be the tangent bundle of a manifold  $Y$ .
- Let  $L = \{(l, z) \mid l \in \mathbb{RP}^\infty, z \in l\}$ . Let  $H : L \rightarrow \mathbb{RP}^\infty$  denote the tautological/canonical/Hopf line bundle over  $\mathbb{RP}^\infty$ . The restriction over  $\mathbb{RP}^n$  is denoted by  $H_n$ .
- $tot(\cdot)$  means the total space of a bundle
- $Prin(X, G)$  denotes the set of principal  $G$ -bundles over  $X$
- The important constructions are colored in [blue](#)

## 1 Kervaire Invariant

### 1.1 Arf Invariant

This section is a summary of [5].

Naively, a quadratic form is a linear  $k$ -function over  $k$ -vector space  $q : V \rightarrow k$  for a given field  $k$  such that  $q(tx) = t^2q(x)$  for any  $t \in k$  and  $x \in V$ .

**Problem 1.1.** When  $k = \mathbb{Z}_2$ , the condition should be  $q(0x) = 0$  and  $q(1x) = q(x)$  for all  $x \in V$ , which is just  $q(0) = 0$ . This condition appears too weak, and we must seek a new definition for the quadratic form.

This problem leads to the definition of a quadratic form on a  $\mathbb{Z}_2$ -vector space.

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**Definition 1.2.** Let  $V$  be a  $\mathbb{Z}_2$ -vector space. A function  $q : V \rightarrow \mathbb{Z}_2$  is said to be a quadratic form if  $I : V \times V \rightarrow \mathbb{Z}_2$  defined by  $I(x, y) = q(x+y) - q(x) - q(y)$  is a bilinear map. In this case, we call  $q$  a quadratic refinement of the bilinear form  $I$ , and  $I$  is the associated bilinear form of  $q$ . We say this quadratic form is nondegenerate if the associated bilinear form  $I$  is nondegenerate, i.e.,  $V \rightarrow \text{Hom}(V, \mathbb{Z}_2)$ ,  $v \mapsto I(v, -)$  is an isomorphism.

**Remark 1.3.** The quadratic refinement of the bilinear form is not unique. An example is in **Construction 1.9**.  $\square$

**Definition 1.4.** Let  $V$  be a  $\mathbb{Z}_2$ -vector space and  $I$  be a bilinear form on  $V$  with quadratic refinement  $q : V \rightarrow \mathbb{Z}_2$ . If  $V$  has a basis  $\{a_i, b_i | i = 1, 2, \dots, s\}$  fulfilling  $I(a_i, a_j) = I(b_i, b_j) = 0$  and  $I(a_i, b_j) = I(a_j, b_i) = \delta_{ij}$ , then we call this basis a symplectic basis for the bilinear form  $I$ .

**Fact 1.5.** If a  $\mathbb{Z}_2$ -vector space admits a nondegenerate quadratic form  $q : V \rightarrow \mathbb{Z}_2$ , then a symplectic basis exists. In particular,  $\dim V$  is even.

**Definition 1.6.** Let  $q : V \rightarrow \mathbb{Z}_2$  be a nondegenerate quadratic form on  $\mathbb{Z}_2$ -vector space  $V$ . By **Fact 1.5**, there exists a symplectic basis  $\{a_i, b_i | i = 1, 2, \dots, s\}$  for the associated bilinear form  $I$ . The Arf-invariant of  $q$  is defined as

$$\text{Arf}(q) = \sum_{i=1}^s q(a_i)q(b_i).$$

For the remaining part, let us focus on what the Arf invariant characterizes.

**Definition 1.7.** Let  $q, q' : V \rightarrow \mathbb{Z}_2$  be two quadratic forms. We say that  $q$  is *equivalent* to  $q'$ , denoted by  $q \sim q'$ , if there exists an automorphism  $f : V \rightarrow V$  such that  $q \circ f = q'$ .

Clearly, this defines an equivalence relation. The following fact shows that the Arf invariant is invariant under this equivalence relation.

**Fact 1.8.** The Arf invariant is a function on the equivalence classes of quadratic forms.

**Construction 1.9.** (For a bilinear map, its quadratic refinement is not unique.)

Let  $W = \mathbb{Z}_2 a_1 \oplus \mathbb{Z}_2 b_1$  be a  $\mathbb{Z}_2$ -vector space with basis  $\{a_1, b_1\}$ . A bilinear map on  $V \times V$  is determined by

$$I(a_1, a_1), \quad I(a_1, b_1), \quad I(b_1, a_1), \quad I(b_1, b_1).$$

A bilinear form admitting a quadratic refinement should satisfy

$$I(a_1, a_1) = I(b_1, b_1) = 0, \quad I(a_1, b_1) = I(b_1, a_1).$$

If we further require that this bilinear form be nondegenerate, then the only possibility is

$$I(a_1, a_1) = I(b_1, b_1) = 0, \quad I(a_1, b_1) = I(b_1, a_1) = 1.$$

As a set,  $W = \{a_1, b_1, a_1 + b_1, 0\}$ . Therefore, a quadratic refinement of  $q$  is determined by the values of  $q(a_1)$ ,  $q(b_1)$ , and  $q(a_1 + b_1)$ .

Define two quadratic forms as follows:

$$q_0(a_1) = q_0(b_1) = 0, q_0(a_1 + b_1) = 1$$

and

$$q_1(a_1) = q_1(b_1) = q_1(a_1 + b_1) = 1$$

We can easily check that both are quadratic refinements of  $I$ . Moreover,  $q_0$  is not equivalent to  $q_1$ , since  $q_0$  sends most elements to 0 while  $q_1$  sends most elements to 1. An automorphism on a vector space does not alter the proportions of the values. Therefore, equivalent quadratic forms map elements to 0 and 1 in the same ratio.  $\square$

**Construction 1.10.** Let  $U$  be a  $\mathbb{Z}_2$ -vector space that admits a nondegenerate quadratic form, and  $\dim U = 2m$ ,  $m \in \mathbb{N}$ . By **Fact 1.5**,  $U$  has a symplectic basis

$$\{a_i, b_i \mid i = 1, 2, \dots, m\}$$

for the associated bilinear form  $I$ .

Then

$$U = (\mathbb{Z}_2 a_1 \oplus \mathbb{Z}_2 b_1) \oplus (\mathbb{Z}_2 a_2 \oplus \mathbb{Z}_2 b_2) \oplus \dots \oplus (\mathbb{Z}_2 a_m \oplus \mathbb{Z}_2 b_m) = W_1 \oplus W_2 \oplus \dots \oplus W_m,$$

where  $W_i = \mathbb{Z}_2 a_i \oplus \mathbb{Z}_2 b_i$ . Each  $W_i$  is a 2-dimensional vector space, so  $W_i$  has two quadratic forms  $q_0, q_1$  in **Construction 1.9**.

We have quadratic forms:

$$mq_0 := q_0 \oplus q_0 \oplus \dots \oplus q_0 : U = W_1 \oplus W_2 \oplus \dots \oplus W_m \rightarrow \mathbb{Z}_2,$$

and

$$q_1 + (m-1)q_0 := q_1 \oplus q_0 \oplus \dots \oplus q_0 : W_1 \oplus \dots \oplus W_m \rightarrow \mathbb{Z}_2.$$

These two quadratic forms are the standard forms for nondegenerate quadratic forms.  $\square$

**Fact 1.11.** (Classification by the Arf invariant) The Arf invariant classifies all nondegenerate quadratic forms. More explicitly,

$$\text{Arf}(q) = 0 \iff q \text{ maps the majority of elements to } 0 \iff q \sim mq_0,$$

$$\text{Arf}(q) = 1 \iff q \text{ maps the majority of elements to } 1 \iff q \sim q_1 + (m-1)q_0.$$

## 1.2 Kervaire Invariant

In this section,  $H^i(M) := H^i(M, \mathbb{Z})$ .

Let  $(M, F)$  be a framed manifold. Using framing, we can define a quadratic form  $\phi : H^n(M; \mathbb{Z}_2) \rightarrow \mathbb{Z}_2$ . Let us only define the Kervaire invariant for a special case. The reference here is [4].

Let  $M^{10}$  be a closed triangulable 4-connected manifold. Let  $\Omega := \Omega S^6$  be the loop space of  $S^6$ .

**Fact 1.12.**  $H^5(\Omega S^6) = \mathbb{Z}e_1$ ,  $H^{10}(\Omega S^6) = \mathbb{Z}e_2$ . Let  $\pi : \Omega \times \Omega \rightarrow \Omega$  be the product of loops, then  $\pi^*(e_1) = e_1 \otimes 1 + 1 \otimes e_1$  and  $\pi^*(e_2) = e_2 \otimes 1 + 1 \otimes e_2 + e_1 \otimes e_1$ .

**Fact 1.13.** Let  $X$  be any element in  $H^5(M)$ . There exists a map  $f : M \rightarrow \Omega$  such that  $f^*(e_1) = X$ .

**Construction 1.14.** Let  $X \in H^5(M)$ , then there exists a map  $f_X : M \rightarrow \Omega$  such that  $f_X^*(e_1) = X$ . Let  $u_2 \in H^{10}(\Omega; \mathbb{Z}_2)$  be the reduction modulo 2 of  $e_2 \in H^{10}(\Omega)$  and  $[M]$  be the generator of  $H_{10}(M; \mathbb{Z})$ . Then we define a map  $\phi_0 : H^5(M) \rightarrow \mathbb{Z}_2$  by  $\phi_0(X) = f_X^*(u_2)[M]$ .  $\square$

**Remark 1.15.**  $\phi_0(X)$  does not depend on the choice of  $f_X : M \rightarrow \Omega$ .  $\square$

**Fact 1.16.**  $\phi_0(X + Y) = \phi_0(X) + \phi_0(Y) + X \cdot Y$  for  $X, Y \in H^5(M)$ . Then  $\phi_0$  induces a map  $\phi : H^5(M; \mathbb{Z}_2) \rightarrow \mathbb{Z}_2$  also satisfying  $\phi(x + y) = \phi(x) + \phi(y) + x \cdot y$  for  $x, y \in H^5(M; \mathbb{Z}_2)$ .

Since  $I(x, y) = \phi(x + y) - \phi(x) - \phi(y) = x \cdot y$  is bilinear, so  $\phi$  is a quadratic form.

**Definition 1.17.** Define the Kervaire invariant by  $K(M, F) := \text{Arf}(\phi)$ .

**Remark 1.18.**  $K(M, F) = 1$  if and only if  $\phi$  sends the majority of elements of  $H^5(M)$  to  $1 \in \mathbb{Z}_2$ .  $\square$

## 2 Construction in Dimension 30

### 2.1 Construction of extended power

If a framed manifold  $M$  has Kervaire invariant one, then  $\dim M = 2, 6, 14, 30, 62, 126$ .

**Example 2.1.**  $S^1 \times S^1$ ,  $S^3 \times S^3$ ,  $S^7 \times S^7$  can be framed to have Kervaire invariant one.

This section introduces a construction in dimension 30 by [3]. The construction is of the form known as an extended power.

**Definition 2.2.** Let  $Y$  be a manifold with  $G \subset \Sigma_t$  acting on it freely. Let  $N$  be a topological space and  $G$  acts on  $N^t$  by permutation. Then we can construct

$$Y_G(N) := Y \times_G N^t,$$

which is called the [extended power](#).

**Remark 2.3.** Here is an advantage of considering extended power. That is, the dimension of extended power is computable:

$$\dim(Y \times_G N^t) = \dim Y + t \dim N.$$

$G$  acts on  $Y$  and  $N^t$  freely. So  $G$  acts freely on  $Y \times N^t$ . Review Proposition 1.40 in [2], condition  $(\star)$  automatically holds for the finite group  $G$ . Hence,

$Y \times N^t \rightarrow Y \times_G N^t$  is a normal covering space. By the property of covering space,

$$\dim(Y \times_G N^t) = \dim(Y \times N^t) = \dim Y + t \dim N.$$

□

**Fact 2.4.** Let  $G$  and  $H$  be any two Lie groups.  $EH \times_H BG$  is a model for  $B(G \rtimes H)$ .

**Construction 2.5.** Let  $\overline{X} := \mathbb{RP}^2 \# T^2$ . Let  $G = \Sigma_2 \wr \Sigma_2 = (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2 \subset \Sigma_4$ .

Let  $a : \mathbb{RP}^2 \subset B\Sigma_2 \xrightarrow{i} B\Sigma_2 \times B\Sigma_2 \xrightarrow{j} E\Sigma_2 \times_{\Sigma_2} (B\Sigma_2 \times B\Sigma_2) = BG$ , where  $j$  is inclusion. The last equation holds by **Fact 2.4**.

Let  $b : T^2 = S^1 \times S^1 \hookrightarrow B\Sigma_2 \times B\Sigma_2 \xrightarrow{1 \times_{\Sigma_2} \Delta} E\Sigma_2 \times_{\Sigma_2} (B\Sigma_2 \times B\Sigma_2) =: BG$ . Here  $1 \times_{\Sigma_2} \Delta$  is constructed as following:

- Define  $E\Sigma_2 \times B\Sigma_2 \xrightarrow{1 \times \Delta} E\Sigma_2 \times (B\Sigma_2 \times B\Sigma_2)$ . Both sides have a  $\Sigma_2$ -action as follows. For  $(a, b) \in E\Sigma_2 \times B\Sigma_2$ ,  $(c, (d, e)) \in E\Sigma_2 \times (B\Sigma_2 \times B\Sigma_2)$ , we have  $\sigma \cdot (a, b) = (\sigma \cdot a, b)$  and  $\sigma(c, (d, e)) = (\sigma \cdot c, (e, d))$ .
- The following diagram shows  $1 \times \Delta$  is  $\Sigma_2$ -equivalence:

$$\begin{array}{ccc} (a, b) & \xrightarrow{\quad} & (a, (b, b)) \\ \downarrow & & \downarrow \\ (\sigma \cdot a, b) & \xrightarrow{\quad} & (\sigma \cdot a, (b, b)) \end{array}$$

$$\begin{array}{ccc} E\Sigma_2 \times B\Sigma_2 & \longrightarrow & E\Sigma_2 \times (B\Sigma_2 \times B\Sigma_2) \\ \sigma \cdot \downarrow & & \downarrow \sigma \cdot \\ E\Sigma_2 \times B\Sigma_2 & \longrightarrow & E\Sigma_2 \times (B\Sigma_2 \times B\Sigma_2) \end{array}$$

So  $1 \times \Delta$  is  $\Sigma_2$ -equivalence. Hence, we can obtain a map

$$1 \times_{\Sigma_2} \Delta : (E\Sigma_2 \times B\Sigma_2) / \Sigma_2 \rightarrow (E\Sigma_2 \times (B\Sigma_2 \times B\Sigma_2)) / \Sigma_2$$

which is

$$1 \times_{\Sigma_2} \Delta : B\Sigma_2 \times B\Sigma_2 \rightarrow E\Sigma_2 \times_{\Sigma_2} (B\Sigma_2 \times B\Sigma_2).$$

Then we can construct  $c : \overline{X} = \mathbb{RP}^2 \# T^2 \xrightarrow{\text{contraction}} \mathbb{RP}^2 \vee T^2 \xrightarrow{a \vee b} BG$ . Let  $X \rightarrow \overline{X}$  be the principal  $G$ -bundle classified by  $c$ . □

We consider the extended power  $X_G(S^7)$ . Note that

$$\dim X_G(S^7) = \dim X + 4 \dim S^7 = 2 + 4 \times 7 = 30$$

by **Remark 2.10**.

**Remark 2.6.** Here are some reasons for choosing such an  $\overline{X}$ . One reason is that  $H^*(\overline{X})$  is easily computed. Besides, the goal of finding  $\overline{X} \rightarrow BG$  can be divided into constructing  $\mathbb{RP}^2 \rightarrow BG$  and  $T^2 \rightarrow BG$ . □

## 2.2 Show $X_G(S^7)$ is framed

Let  $\xi : X_G(\mathbb{R}) \rightarrow \bar{X}$  be the 4-dimensional bundle. Then by **Theorem A** of [3], to show that  $X_G(S^7)$  is framed is equivalent to showing that  $\tau\bar{X} + 7\xi$  is stably trivial. Since all stable bundles over a surface are classified by their Stiefel–Whitney classes, it suffices to show

$$w(\tau\bar{X} + 7\xi) = w(\tau\bar{X})w(\xi)^7 = 1.$$

$H^1(\bar{X}) = H^1(\mathbb{RP}^2) + H^1(T^2) = \mathbb{Z}_2u + \mathbb{Z}_2x_1 + \mathbb{Z}_2x_2$ , where  $u$  is the generator coming from  $H^1(\mathbb{RP}^2)$ , and  $x_1, x_2$  are the generators coming from  $H^1(T^2)$ .  $H^2(\bar{X}) = \mathbb{Z}_2u^2$ . Note that  $u^2 = x_1x_2 \neq 0$ ,  $x_1^2 = x_2^2 = ux_1 = ux_2 = u^3 = 0$ .

By [1]  $w(\tau\bar{X}) = w(\tau\mathbb{RP}^2 \# T^2) = w(\tau\mathbb{RP}^2) + w(\tau T^2) - 1 = 1 + u + u^2 + 1 - 1 = 1 + u + u^2$ .

Let  $r : G \rightarrow O(4)$  be the permutation representation. Let  $\rho$  be the bundle classified by  $Br$ .

**Remark 2.7.** Clearly, a normal bundle of  $T^2$  is trivial. So  $w(\nu T^2) = 1$ . By  $w(\tau T^2 \oplus \nu T^2) = w(\tau T^2)w(\nu T^2) = 1$ , we get  $w(\tau T^2) = 1$ . The computation of  $w(\tau\mathbb{RP}^2)$  is more complicated. Note that  $\tau\mathbb{RP}^2 \oplus \varepsilon^1 = \oplus^3 H_2$  where  $H_2$  is the tautological bundle over  $\mathbb{RP}^2$ . So  $w(\tau\mathbb{RP}^2) = w(H_2)^3 = (1 + u)^3 = 1 + u + u^2$ . For more details, see [5].  $\square$

**Claim 2.8.**  $\xi = c^*\rho$ .

$\rho$  is the bundle classified by  $Br$ , so by **Remark 2.10**,  $\rho : EG \times_{G,r} O(4) \rightarrow BG$ . Then  $c^*\rho = X \times_{G,r} O(4) \rightarrow X$  by **Remark 2.11**, which is a principal  $O(4)$ -bundle. There exists a corresponding  $O(4)$ -bundle with fiber  $\mathbb{R}^4$ , which has total space  $(X \times_{G,r} O(4)) \times_{O(4)} \mathbb{R}^4 = X \times_{G,r} (O(4) \times_{O(4)} \mathbb{R}^4) = X \times_{G,r} \mathbb{R}^4 = X_G(\mathbb{R})$ . Hence  $c^*\rho : X_G(\mathbb{R}) \rightarrow X$  is  $\xi$ .

**Remark 2.9.** Let  $p \in P$ ,  $g \in G$ ,  $h \in H$ . For  $\theta : H \rightarrow G$ , we can define the balanced product  $P \times_{H,\theta} G := P \times G / \sim$ , where  $(ph, g) \sim (p, hg)$ . Equivalently,  $P \times_{H,\theta} G = (P \times G) / H$ , where  $h(p, g) = (ph^{-1}, hg)$ . Balanced product satisfies associativity and  $W \times_G G \simeq G$  for any group  $G$  and  $G$ -space  $W$ .  $\square$

**Remark 2.10.** Let us consider a more general question: Let  $H, G$  be any groups,  $X$  be any topological space. Given a group homomorphism  $\theta : H \rightarrow G$ , what's  $B\theta$ ? Here provides a construction of the map  $B\theta$  (and this construction works).

Goal: construct an element in  $\text{Hom}(BH, BG)$ . By Yoneda lemma, we need to construct an element in  $\text{Hom}([- , BH], [- , BG])$ , i.e., we need to construct  $\phi_X : [X, BH] \rightarrow [X, BG]$  for any  $X$ . By isomorphisms  $\text{Prin}(X, H) \simeq [X, BH]$  and  $\text{Prin}(X, G) \simeq [X, BG]$ , we need to construct the map induced by  $\phi, \bar{\phi}_X : \text{Prin}(X, H) \rightarrow \text{Prin}(X, G)$ . Given a principal  $H$ -bundle  $P \rightarrow X$ , we want to construct a principal  $G$ -bundle, using data  $P \rightarrow X$  and  $\theta : H \rightarrow G$ . Note that  $\theta : H \rightarrow G$  makes  $H$  acting left on  $G$ . So we can form an  $H$ -bundle with fiber  $G$ :  $P \times_{H,\theta} G \rightarrow X$ . Clearly we can view  $[P \times_{H,\theta} G \rightarrow X]$  as a  $G$ -bundle with  $G$

action on fiber  $G$  by multiplication. So  $[P \times_{H,\theta} G \rightarrow X] \in B(X, G, G, m_G) =: \text{Prin}(X, G)$ .

By Yoneda lemma,  $\text{Hom}([- , BH], [- , BG]) \rightarrow \text{Hom}(BH, BG)$  defined by  $\Psi \mapsto \Psi_{BH}(\text{id})$  is an isomorphism. Let  $\Psi = \phi$  and define  $B\theta := \phi_{BH}(\text{id})$ , the image of  $\phi$  under Yoneda map. Since  $\bar{\phi}_X$  is induced by  $\phi_X$ , we have commutative diagram:

$$\begin{array}{ccc}
[BH, BH] & \xrightarrow{\phi_{BH}} & [BH, BG] \\
\downarrow \simeq & & \downarrow \simeq \\
\text{Prin}(BH, H) & \xrightarrow{\bar{\phi}_{BH}} & \text{Prin}(BH, G) \\
\\ 
id & \longmapsto & \phi_{BH}(id) =: B\theta \\
\downarrow & & \downarrow \\
id^*EH = [EH \rightarrow BH] & \longmapsto & [EH \times_{H,\theta} G \rightarrow BH]
\end{array}$$

From this diagram, even we do not know what  $B\theta$  is, we can know  $B\theta$  classifies principal  $G$ -bundle  $EH \times_{H,\theta} G \rightarrow BH$ .  $\square$

**Remark 2.11.** Consider this problem: Let  $G$  be any group,  $X$  be any topological space, and  $\pi : E \rightarrow E/G$  be a bundle. The pullback of  $\pi$  along  $c : X \rightarrow E/G$  is  $M$ . Let  $L$  be the pullback of  $E \times_G F \rightarrow E/G$  along  $c$ . Then what is the relationship between  $L$  and  $M$ ?

By construction,

$$\begin{aligned}
L &= X \times_{E/G} (E \times_G F) \\
&= \{(x, e, f) \mid x \in X, e \in E, f \in F, c(x) = [e]_G\} / (x, e, f) \sim (x, eg, g^{-1}f),
\end{aligned}$$

for any  $g \in G$ .

$$\begin{aligned}
M \times_G F &= (X \times_{E/G} E) \times_G F \\
&= \{(x, e, f) \mid x \in X, e \in E, f \in F, c(x) = [e]_G\} / (x, e, f) \sim (x, eg, g^{-1}f)
\end{aligned}$$

for any  $g \in G$ .

So  $L = M \times_G F$ .  $\square$

Since  $\xi = c^*\rho$ , we need to compute  $w(c^*\rho)$ . To compute  $w(c^*\rho)$ , we need to compute  $a^*\rho$  and  $b^*\rho$ .

$G = \Sigma_2 \wr \Sigma_2 \subset \Sigma_4$  is the subgroup generated by (12), (34), (13)(24) by **Remark 2.12**. Let  $i_1 : \Sigma_2 \times \Sigma_2 \rightarrow G$  be the inclusion of the subgroup generated by (12) and (34). Let  $i_2 : \Sigma_2 \times \Sigma_2 \rightarrow G$  be the inclusion of the subgroup generated by (12)(34) and (13)(24). Let  $s : \Sigma_2 \rightarrow O(2)$  be a permutation representation.

**Remark 2.12.**  $G = (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2$ , where the multiplication is

$$((h_0, h_1), x) \cdot ((l_0, l_1), y) = ((y \cdot (h_0, h_1))(l_0, l_1), xy)$$

and

$$\begin{cases} \sigma(h_0, h_1) &= (h_1, h_0), \\ 1(h_0, h_1) &= (h_0, h_1) \end{cases}$$

$G$  is isomorphic to the subgroup in  $\Sigma_4$  generated by (12), (34), (13)(24). The correspondence is as follows:

|                              |          |
|------------------------------|----------|
| $((1, 1), 1)$                | 1        |
| $((\sigma, 1), 1)$           | (12)     |
| $((1, \sigma), 1)$           | (34)     |
| $((\sigma, \sigma), 1)$      | (12)(34) |
| $((1, 1), \sigma)$           | (13)(24) |
| $((\sigma, 1), \sigma)$      | (1423)   |
| $((1, \sigma), \sigma)$      | (1324)   |
| $((\sigma, \sigma), \sigma)$ | (14)(23) |

Table 1: Correspondance

□

**Claim 2.13.**  $Bs$  classified the bundle  $H + \varepsilon^1$ .

By **Remark 2.10**,  $Bs$  induces map classifies the principal  $O(2)$ -bundle  $E\Sigma_2 \times_{\Sigma_2} O(2) \rightarrow B\Sigma_2$ . This principal  $O(2)$ -bundle corresponds to an element in  $B(B\Sigma_2, O(2), \mathbb{R}^2, \rho)$ , where  $\rho : O(2) \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ ,  $(M, x) \mapsto M \cdot x$ . The associated  $O(2)$ -bundle with fiber  $\mathbb{R}^2$  is  $E\Sigma_2 \times_{\Sigma_2} O(2) \times_{O(2)} \mathbb{R}^2 = E\Sigma_2 \times_{\Sigma_2} \mathbb{R}^2 \rightarrow B\Sigma_2$ .

Write  $u = [\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}] \in \mathbb{R}^2$  and  $v = [\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}] \in \mathbb{R}^2$ . We observe that  $\sigma u = u$  and  $\sigma v = -v$ . (Here  $u$  and  $v$  are more symmetric than a standard basis.)

Check that the total spaces are the same:

- $E\Sigma_2 \times_{\Sigma_2} \mathbb{R}^2 = \{(z, xu + yv) \mid z \in E\Sigma_2 = S^\infty, x, y \in \mathbb{R}\} / \sim$ , where  $(z, xu + yv) \sim (z\sigma^{-1}, \sigma(xu + yv)) = (-z, xu - yv)$ . Let  $w = z \cdot y$  and  $x = t$ , then  $E\Sigma_2 \times_{\Sigma_2} \mathbb{R}^2 = \{(w, t) \mid w \in \mathbb{R}^\infty, t \in \mathbb{R}\}$ .
- $\text{tot}(H + \varepsilon^1) = \{((l, w), t) \mid l \in \mathbb{RP}^\infty, w \in l \subset \mathbb{R}^\infty, t \in \mathbb{R}\} = \{(w, t) \mid w \in \mathbb{R}^\infty, t \in \mathbb{R}\} = E\Sigma_2 \times_{\Sigma_2} \mathbb{R}^2$ .

Check that the bundle maps are the same:

- $H + \varepsilon^1 : \text{tot}(H + \varepsilon^1) \rightarrow \mathbb{RP}^\infty$  is defined by  $(w, t) \mapsto [w]$ , where  $[w]$  denotes the line  $w$  lies in.
- $E\Sigma_2 \times_{\Sigma_2} \mathbb{R}^2 \rightarrow B\Sigma_2$  is defined by  $(z \cdot y, x) \mapsto [z] = [z \cdot y]$ , which is  $(w, t) \mapsto [w]$ .

Therefore, the two bundles are exactly the same.

**Claim 2.14.**  $ri_1 = s \times s$ .

$\Sigma_2 \times \Sigma_2 \xrightarrow{s \times s} O(2) \times O(2) \subset O(4)$  is defined by

$$(\sigma, 0) \mapsto \begin{bmatrix} & 1 & & \\ 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad (0, \sigma) \mapsto \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{bmatrix}.$$

and  $\Sigma_2 \times \Sigma_2 \xrightarrow{i_1} G \xrightarrow{r} O(4)$  is defined by

$$(\sigma, 0) \mapsto (12) \mapsto \begin{bmatrix} & 1 & & \\ 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad (0, \sigma) \mapsto (34) \mapsto \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{bmatrix}.$$

So  $s \times s = ri_1$ .

**Claim 2.15.**  $s \otimes s = ri_2$ .

$\Sigma_2 \times \Sigma_2 \xrightarrow{i_2} G \xrightarrow{r} O(4)$  is defined by

$$(\sigma, 1) \mapsto (12)(34) \mapsto \begin{bmatrix} & 1 & & \\ 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \quad (1, \sigma) \mapsto (13)(24) \mapsto \begin{bmatrix} & & 1 & \\ 1 & & & \\ & 1 & & \\ & & & 1 \end{bmatrix}.$$

Since  $s : \Sigma_2 \rightarrow O(2)$  is defined by  $\sigma \mapsto \begin{bmatrix} & 1 \\ 1 & \end{bmatrix}$ ,  $1 \mapsto \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$ ,

then  $s \otimes s : \Sigma_2 \otimes \Sigma_2 \rightarrow O(2) \otimes O(2)$  satisfies

$$\begin{aligned} s(\sigma \otimes 1) &= \sigma \otimes s1 = \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \otimes \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} = \begin{bmatrix} & 1 & & \\ 1 & & & \\ & & 1 & \\ & & & 1 \end{bmatrix}, \\ s(1 \otimes \sigma) &= s1 \otimes s\sigma = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} \otimes \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} = \begin{bmatrix} & & 1 & \\ 1 & & & \\ & 1 & & \\ & & & 1 \end{bmatrix}. \end{aligned}$$

So  $s \otimes s = ri_2$ .

**Claim 2.16.**  $j = Bi_1$ ;  $1 \times_{\Sigma_2} \Delta = Bi_2$ .

By **Remark 2.12**  $i_1, i_2$  are actually:

$$\begin{aligned} i_1 : \Sigma_2 \times \Sigma_2 &\rightarrow (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2, & (\sigma, 1) &\mapsto ((\sigma, 1), 1), & (1, \sigma) &\mapsto ((1, \sigma), 1), \\ i_2 : \Sigma_2 \times \Sigma_2 &\rightarrow (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2, & (\sigma, 1) &\mapsto ((\sigma, \sigma), 1), & (1, \sigma) &\mapsto ((1, 1), \sigma). \end{aligned}$$

So  $i_1$  can also be written as  $\Sigma_2 \times \Sigma_2 \cong (\Sigma_2 \times \Sigma_2) \rtimes 1 \hookrightarrow (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2$ . Apply functor  $B$  and use **Fact 2.4**, we obtain

$$B(\Sigma_2 \times \Sigma_2) \xrightarrow{\cong} B(\Sigma_2 \times \Sigma_2) \times_{\{1\}} E_{\{1\}} \longrightarrow B(\Sigma_2 \times \Sigma_2) \times_{\Sigma_2} E\Sigma_2.$$

Then  $Bi_1 = j$ .

Consider  $\Sigma_2 \times \Sigma_2 \xrightarrow{\Delta \rtimes 1} (\Sigma_2 \times \Sigma_2) \rtimes \Sigma_2$  defined by  $(a, b) \mapsto ((a, a), b)$ , which is a group homomorphism since  $b \cdot (a, a) = (a, a)$  for any  $b \in \Sigma_2$ . We observe that  $i_2 = \Delta \rtimes 1$ .

Apply functor  $B$  and use **Fact 2.4**, we obtain

$$B\Sigma_2 \times B\Sigma_2 \xrightarrow{\Delta \times \Sigma_2 1} B\Sigma_2 \times B\Sigma_2 \times_{\Sigma_2} E\Sigma_2$$

**Remark 2.17.** We use  $G$  being a subgroup of  $\Sigma_4$  (so we have group representation) to prove  $ri_1 = s \times s$  and  $ri_2 = s \otimes s$ . We use  $G = \Sigma_2 \wr \Sigma_2$  (so  $BG = E\Sigma_2 \times_{\Sigma_2} (B\Sigma_2 \times B\Sigma_2)$ ) in computing  $j = Bi_1$ ,  $1 \times_{\Sigma_2} \Delta = Bi_2$ .  $\square$

Consider the commutative diagram:

$$\begin{array}{ccccc} \text{tot}(j^*\rho) & \longrightarrow & \text{tot}(\rho) & \longrightarrow & EO(4) \\ j^*\rho \downarrow & & \rho \downarrow & & \downarrow \\ B(\Sigma_2 \times \Sigma_2) & \xrightarrow{Bi_1=j} & BG & \xrightarrow{Br} & BO(4) \\ & \searrow & & \nearrow & \\ & B(ri_1)=B(s \times s)=Bs \times Bs & & & \end{array}$$

So by **Claim 2.13**,  $j^*\rho = (H + \varepsilon^1) \times (H + \varepsilon^1)$ .

**Remark 2.18.** Consider a more general problem: Let  $A \subset B_1$ ,  $\pi_1 : E_1 \rightarrow B_1$ ,  $\pi_2 : E_2 \rightarrow B_2$  be two vector bundles and  $E_2 \rightarrow B_2$  has rank  $n$ . Then we have pullback:

$$\begin{array}{ccc} E_1|_A \times \epsilon^n & \longrightarrow & E_1 \times E_2 \\ \downarrow & & \downarrow \pi_1 \times \pi_2 \\ A \simeq A \times \{pt\} & \longrightarrow & B_1 \times B_2 \end{array}$$

$\square$

By **Remark 2.18**, the bundle over  $\mathbb{RP}^2$  is  $(H + \varepsilon^1)|_{\mathbb{RP}^2} + \varepsilon^2 = H_2 + \varepsilon^1 + \varepsilon^2 = H_2 + \varepsilon^3$ , i.e.,  $a^*\rho = H_2 + \varepsilon^3$ .

Consider a commutative diagram:

$$\begin{array}{ccccc} \text{tot}(Bi_2^*\rho) & \longrightarrow & \text{tot}(\rho) & \longrightarrow & EO(4) \\ Bi_2^*\rho \downarrow & & \rho \downarrow & & \downarrow \\ B(\Sigma_2 \otimes \Sigma_2) & \xrightarrow{Bi_2} & BG & \xrightarrow{Br} & BO(4) \\ & \searrow & & \nearrow & \\ & Bs \otimes s & & & \end{array}$$

So  $Bi_2^*\rho = (H + \varepsilon^1) \otimes (H + \varepsilon^1)$ . Similarly as **Remark 2.18**,  $b^*\rho = (H_1 + \varepsilon) \otimes (H_1 + \varepsilon) = H_1 \otimes H_1 + H_1 \otimes \varepsilon^1 + \varepsilon^1 \otimes H_1 + \varepsilon^1 \otimes \varepsilon^1$ .

**Remark 2.19.** Here is a commutative diagram that helps readers understand relations.

$$\begin{array}{ccccc}
(H_1 + \varepsilon^1) \otimes (H_1 + \varepsilon^1) & \longrightarrow & (H + \varepsilon^1) \otimes (H + \varepsilon^1) & \longrightarrow & EO(4) \\
\downarrow & & \downarrow & & \downarrow \\
S^1 \times S^1 = \mathbb{RP}^1 \times \mathbb{RP}^1 & \longrightarrow & B(\Sigma_2 \otimes \Sigma_2) & \xrightarrow{Bs \otimes s} & BO(4)
\end{array}$$

□

**Claim 2.20.**  $w(H_1) = 1 + x$ .

Since  $H_1$  is a line bundle, we only need to compute  $w_1(H_1)$ . Note that  $H^*(\mathbb{RP}^1) = \mathbb{Z}_2[x]/x^2$ ,  $H^*(\mathbb{RP}^\infty) = \mathbb{Z}_2[x]$ . Then  $\iota : \mathbb{RP}^1 \rightarrow \mathbb{RP}^\infty$  induces  $\iota^* : H^*(\mathbb{RP}^\infty) \rightarrow H^*(\mathbb{RP}^1)$ . By axiom of Stiefel-Whitney class,  $w_1(H) \neq 0$ , so  $w_1(H) = x$ . Then  $w_1(\iota^*H) = \iota^*(w_1(H)) = \iota^*x = x \in H^1(\mathbb{RP}^1)$ . So  $w(H_1) = 1 + x$ .

Then we can compute:

$$\begin{aligned}
w(a^*\rho) &= w(H_2)w(\varepsilon^3) = w(H_2) = 1 + u, \\
w(b^*\rho) &= w(H_1 \otimes H_1)w(H_1 \otimes \varepsilon^1)w(\varepsilon^1 \otimes H_1)w(\varepsilon^1 \otimes \varepsilon^1).
\end{aligned}$$

Since they are all line bundles, we only need to compute the first Stiefel-Whitney class.

**Fact 2.21.** Let  $L_1, L_2$  be real line bundles over a paracompact space  $B$ . Then  $w_1(L_1 \otimes L_2) = w_1(L_1) + w_1(L_2)$ .

Hence,

$$\begin{aligned}
w_1(H_1 \otimes H_1) &= w_1(H_1) + w_1(H_1) = x_1 + x_2, \\
w_1(H_1 \otimes \varepsilon^1) &= x_1 + 0, \\
w(\varepsilon^1 \otimes H_1) &= 0 + x_2, \\
w(\varepsilon^1 \otimes \varepsilon^1) &= 0.
\end{aligned}$$

So

$$w(b^*\rho) = (1 + x_1 + x_2)(1 + x_1)(1 + x_2) \cdot 1 = 1 + x_1x_2$$

(using  $x_1^2 = x_2^2 = 0$  and note that we are in  $\mathbb{Z}_2$ -coefficients).

$w_1(\xi) = w_1(c^*\rho) = w_1((a \vee b)^*\rho) = w_1(a^*\rho) + w_1(b^*\rho) = u + x_1x_2 = u + u^2$ , where the third equation holds by **Remark 2.22**.

**Remark 2.22.**  $\iota_1 : P \hookrightarrow P \vee T$  and  $\iota_2 : T \hookrightarrow P \vee T$  induce projection on cohomology  $\iota_1^* : H^i(P \vee T) = H^i(P) \oplus H^i(T) \rightarrow H^i(P)$  and  $\iota_2^* : H^i(P \vee T) =$

$H^i(P) \oplus H^i(T) \rightarrow H^i(T)$ , i.e., for any  $x \in H^i(P \vee T)$ , we have  $x = \iota_1^*x + \iota_2^*x$ . Let  $f = a \vee b$ . Then  $f\iota_1 = a$ ,  $f\iota_2 = b$ . For any  $w_i(f^*\rho) \in H^i(P \vee T)$ , we have

$$\begin{aligned} w_i(f^*\rho) &= \iota_1^*w_i(f^*\rho) + \iota_2^*w_i(f^*\rho) = \iota_1^*f^*w_i(\rho) + \iota_2^*f^*w_i(\rho) \\ &= (f\iota_1)^*w_i(\rho) + (f\iota_2)^*w_i(\rho) = a^*w_i(\rho) + b^*w_i(\rho) = w_i(a^*\rho) + w_i(b^*\rho), \end{aligned}$$

for  $i > 1$ . □

So  $w(\xi) = 1 + u + u^2$ . Then

$$w(\tau\bar{X} + 7\xi) = w(\tau\bar{X})w(\xi)^7 = (1 + u + u^2)^8 = 1.$$

One can find the details of the computation showing that this manifold has Kervaire invariant one in [3].

Constructing an extended power of dimension 30 works, but can this method be used to construct an example in dimension 62? Actually, it is not easy by means of Theorem D in [3].

Let  $G_k := \Sigma_2 \wr \cdots \wr \Sigma_2$  ( $k$ 's  $\Sigma_2$ ). Consider  $Y_{G_k}(S^7)$  with  $\dim Y = d = 2^{l+1} - 2 - 7 \cdot 2^k$ . By **Remark 2.3**,  $\dim Y_{G_k}(S^7) = 2^{l+1} - 2$ .

$2^{l+1} - 2 = 62$  implies  $l = 5$ . With  $\dim Y = 2^6 - 2 - 7 \cdot 2^k \geq 0$ , we have the only possibilities of  $(k, d)$  are  $(0, 55), (1, 48), (2, 34), (3, 6)$ . All cases satisfying  $d > 2$  meeting the condition of Theorem D in [3]. So given any framing of  $S^7$ , and  $\alpha$  a stable trivialization of  $\tau\bar{Y} + 7\xi$ , then  $K(Y_{G_k}(S^7), \alpha_{G_k}(F)) = 0$ . Indeed, replacing  $\alpha_{G_k}(F)$  with another framing can make the Kervaire invariant nontrivial, but it is not easy to find such a framing.

## References

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